

# Mary Pletcher

mary.c.pletcher@gmail.com · <https://marypleletcher.github.io/headinclouds/>

## SYSTEMS AND INTERFACE ENGINEERING

Motivated and curious Systems Engineering Master’s graduate with a passion for tackling challenges and driving impact. Equipped with a foundation in [UI/UX design](#), [human factors engineering](#), [programming fundamentals](#), and [project management](#). Proven track record of independently solving evolving problems and collaborating effectively within teams to deliver exceptional value for users.

## KEY COMPETENCIES

|                            |                               |                            |
|----------------------------|-------------------------------|----------------------------|
| Design Thinking            | Report Writing and Presenting | Usability Testing          |
| UI/UX Design + Prototyping | Project Management            | Survey Design and Analysis |
| Scripting and Automation   | Requirements Gathering        | 3D + VR Prototyping        |

## RELEVANT EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                  |                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Peraton - Systems Engineer</b>                                                                                                                                                                                                                                                                                                                | <b>Summer 2024 - Present</b>     |
| <ul style="list-style-type: none"><li>Led risk reduction data team by organizing and managing modeling and simulation data</li><li>Identified and tracked closure of approximately 30 risks to testing and development</li><li>Implemented digital optimizations and Sources of Truth using Powershell scripting, Jira, and Confluence</li></ul> |                                  |
| <b>Human IMPAC-T Lab - User Researcher and Lab Manager</b>                                                                                                                                                                                                                                                                                       | <b>Spring 2023 - Summer 2024</b> |
| <ul style="list-style-type: none"><li>Spearheaded 3 user experience research projects, balancing high level goals and interaction details</li><li>Taught an undergraduate course on applications of user research/design principles</li><li>Organized best practices and activities through Notion (digital project management tool)</li></ul>   |                                  |
| <b>Virginia Tech Library - Prototyping Studio Consultant</b>                                                                                                                                                                                                                                                                                     | <b>Fall 2021 - Spring 2024</b>   |
| <ul style="list-style-type: none"><li>Leveraged excellent customer service skills to support academic and personal projects</li><li>Developed educational materials for projects and tool training</li><li>Applied user-centered design for a children’s museum interactive installation project</li></ul>                                       |                                  |
| <b>University of Nottingham, UK - Human Factors Researcher</b>                                                                                                                                                                                                                                                                                   | <b>Summer 2023</b>               |
| <ul style="list-style-type: none"><li>Efficiently executed research study within short timeline, demonstrating organization and initiative</li><li>Programmed Unity app for VR/AR experience in simulated autonomous vehicle</li><li>Performing mixed methods data analysis on survey and think aloud protocol data using Python</li></ul>       |                                  |
| <b>VR Immersive Training Project - Undergraduate Research Assistant</b>                                                                                                                                                                                                                                                                          | <b>Spring 2022 - Spring 2023</b> |
| <ul style="list-style-type: none"><li>Collaborated with two students to create a literature review report on AR/VR ergonomic training</li><li>Prototyped VR + motion tracking training app based on application requirements</li></ul>                                                                                                           |                                  |
| <b>Accenture Federal Services - Summer Analyst</b>                                                                                                                                                                                                                                                                                               | <b>Summer 2022</b>               |
| <ul style="list-style-type: none"><li>Collaborated with solutions team on market/technology research presentations and graphics</li><li>Developed 4 decks, 1 one-pager, and supported work and key wins across 6 different proposals</li><li>Initiated and planned summer networking events with other summer interns</li></ul>                  |                                  |

## EDUCATION & CERTIFICATIONS

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|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>M. Eng. in Industrial and Systems Engineering</b><br>Virginia Tech (May 2024)<br>Human Factors Engineering Track<br>GPA: 3.91/4.00 |
| <b>Human-Computer Interaction Graduate Certificate</b><br>Virginia Tech (May 2024)                                                    |
| <b>B. S. in Industrial and Systems Engineering</b><br>Virginia Tech (May 2023)<br>Human Computer Interaction Minor<br>GPA: 3.73/4.00  |

## HIGHLIGHTED PROJECT

**Extended Reality (XR) as a Product at Virginia Tech Graduate Lead (2023-2024)**  
Led a community effort to [determine the business and educational value](#) of XR products. [Collected user requirements and advocated for user needs](#) through [surveys, focus groups, personas, market research, workflow maps, and system maps](#). Continuously reported findings to technology and teaching teams through [update reports, documentation, and presentations](#).